

Md. Abu Tahshin

tahshinsharon@gmail.com | +8801778427031 | tahshinsharon.github.io

Q9XG+43 Dhaka. Rd No 9, Dhaka 1216, East Kazipara, Mirpur, Dhaka

Experience

Shikho Technologies Ltd.

Software Engineer (Back-End) (January 2025 – Present)

Dhaka, Bangladesh

Projects

Dynamic QR-Based FutureBook — *Shikho Technologies, Dhaka, Bangladesh*

- Built a scalable backend system enabling dynamic QR-based integration between physical textbooks and digital learning content, supporting multi-year academic programs, entitlement enforcement, quizzes, analytics, and CMS-controlled QR generation with dynamic updates.
- Designed a **multi-tenant data model** to support multiple academic years and CMS-controlled QR expiry, reuse, and re-issuance rules without downtime.
- Implemented **entitlement checks** and analytics event streaming to track per-student engagement on each QR-linked resource, enabling data-driven content decisions for the CMS team.
- **Technologies:** Golang, GraphQL, REST API, MongoDB, Redis.

Alor-Shikha – Online Educational Quiz Platform — *Shikho Technologies, Dhaka, Bangladesh*

Alor-Shikha.com

- Implemented and maintain **weekly-based exam, scoring, leaderboard, and progress APIs**.
- Maintain **caching strategies** for faster ranking, score retrieval, and **reduced database load**.
- Delivered **sub-100ms leaderboard and score reads under peak submission load**, outperforming the earlier direct-database approach and keeping the ranking view responsive during concurrent traffic spikes.
- **Technologies:** Golang, Redis, GraphQL, REST API, MongoDB.

Focus Hub – Real-Time Analytics Pipeline — *Shikho Technologies, Dhaka, Bangladesh*

- Built a **real-time analytics pipeline** for the **Focus Hub** feature using **database change streams** and a **message broker** to stream document-level changes into a pre-aggregated **PostgreSQL** analytics table.
- Decoupled analytics writes from the hot transactional path, giving near real-time dashboards without adding load or read pressure on the operational database cluster.
- Designed the aggregated schema and consumer workers to be **idempotent and replay-safe**, so lost or duplicated broker messages never corrupt the analytics counters.
- **Technologies:** Golang, MongoDB, NATS, PostgreSQL.

Partner-Campaign & API-Client Platform — *Shikho Technologies, Dhaka, Bangladesh*

- Designed and built an **API-key based access system** allowing external partners to consume **Partner-Campaign** product APIs through client-specific keys generated by our platform.
- Implemented per-client **authentication** so each API key is scoped strictly to that client's allowed campaign resources.
- **Technologies:** Golang, GraphQL, REST API, MongoDB.

CQ Formatter – AI-Assisted Bulk Question Ingestion — *Shikho Technologies, Dhaka, Bangladesh*

- Built an internal tool that automates **bulk upsert of CQ (Creative Question) questions** into our exam system from **image and document uploads**.
- Integrated the **Gemini API** to extract and normalize question content under the hood, so employees only configure the target exam and trigger a bulk insert instead of typing questions manually.

- Reduced manual data-entry effort for the content team by **~70%**, eliminating the biggest bottleneck in shipping new question banks on time.
- **Technologies:** Golang, Gemini API, REST API, MongoDB.

Technical Skills

Languages:	Golang, Python, C, C++
Databases:	PostgreSQL, MySQL, MongoDB, ArangoDB, Elasticsearch
Tools & Platforms:	Git, Docker, Kubernetes, RabbitMQ, Redis, NATS, Grafana
Cloud Platforms:	AWS (EC2, S3, Amplify, DynamoDB, Lambda, IAM, CloudWatch)
Concepts:	Clean Architecture, REST, gRPC, DevOps, GraphQL

Achievements

Solved 1000+ Programming Problems on Online Judges

[Codeforces](#) | [LeetCode](#)

- Cumulative record across Codeforces, LeetCode, and StopStalk.

IEEEExtreme Programming Competition 17.0

Team: [ARSfire](#)

- Rank: **6th in Bangladesh** and **1162 globally** out of **7091 teams**.

ICPC Asia Regional Contest

2023

Team: [JnuLazyPenguin](#)

- Participated in the ACM-ICPC Asia Regional Contest.

JNU Intra University Programming Contest

2023

1st Runner Up

- Ranked **2nd out of 50 participants**.

UITS IUPC

2023

Team: [JnuLazyPenguin](#)

- Ranked **26th**.

Research Experience

Predicting Autism Spectrum Disorder — *Jagannath University*

Feb 2024 – Aug 2024

Dhaka, Bangladesh

- Objective: accurately detect autism in young children using machine learning.
- Achieved **97% accuracy**, **1% higher** than the previous state-of-the-art model.

Extracurricular Activities

Lead Organizer, JNU IUPC — *Jagannath University*

1 March 2024

Dhaka, Bangladesh

[Event Link](#)

- **Problem Setter** for the contest.
- Organized volunteers and distributed tasks.
- Communicated between teachers, students, and alumni.